DBMS
JOINS

INNER JOIN

```
mysql> SELECT Student.Student_ID, Student.Name, Student.Marks, Department.Department_Name, Department.HOD
    -> FROM Student
    -> INNER JOIN Department
    -> ON Student.Dept_ID = Department.Dept_ID;
Empty set (0.01 sec)

mysql> SELECT * FROM Student;
+------------+--------+---------+-------+
| Student_ID | Name   | Dept_ID | Marks |
+------------+--------+---------+-------+
|        201 | Arjun  |      10 |    81 |
|        202 | Sneha  |      20 |    89 |
|        203 | Kiran  |      10 |    74 |
|        204 | Ananya |      30 |    91 |
|        205 | Vikram |      20 |    86 |
+------------+--------+---------+-------+
```

LEFT JOIN

```
mysql> SELECT Student.Student_ID, Student.Name, Student.Marks, Department.Department_Name, Department.HOD
    -> FROM Student
    -> LEFT JOIN Department
    -> ON Student.Dept_ID = Department.Dept_ID;
+------------+--------+-------+-----------------+------------+
| Student_ID | Name   | Marks | Department_Name | HOD        |
+------------+--------+-------+-----------------+------------+
|        201 | Arjun  |    81 | AIML            | Dr. Kumar  |
|        202 | Sneha  |    89 | ISE             | Dr. Reddy  |
|        203 | Kiran  |    74 | AIML            | Dr. Kumar  |
|        204 | Ananya |    91 | EEE             | Dr. Thomas |
|        205 | Vikram |    86 | ISE             | Dr. Reddy  |
+------------+--------+-------+-----------------+------------+
5 rows in set (0.00 sec)
```

RIGHT JOIN

```
mysql> SELECT Student.Student_ID, Student.Name, Student.Marks, Department.Department_Name, Department.HOD
    -> FROM Student
    -> RIGHT JOIN Department
    -> ON Student.Dept_ID = Department.Dept_ID;
+------------+--------+-------+-----------------+------------+
| Student_ID | Name   | Marks | Department_Name | HOD        |
+------------+--------+-------+-----------------+------------+
|        203 | Kiran  |    74 | AIML            | Dr. Kumar  |
|        201 | Arjun  |    81 | AIML            | Dr. Kumar  |
|        205 | Vikram |    86 | ISE             | Dr. Reddy  |
|        202 | Sneha  |    89 | ISE             | Dr. Reddy  |
|        204 | Ananya |    91 | EEE             | Dr. Thomas |
|       NULL | NULL   |  NULL | MECH            | Dr. Patel  |
+------------+--------+-------+-----------------+------------+
6 rows in set (0.01 sec)
```

FULL OUTTER JOIN:

```
mysql> SELECT Student.Student_ID, Student.Name, Student.Marks, Department.Department_Name, Department.HOD
    -> FROM Student
    -> LEFT JOIN Department
    -> ON Student.Dept_ID = Department.Dept_ID
    ->
    -> UNION
    ->
    -> SELECT Student.Student_ID, Student.Name, Student.Marks, Department.Department_Name, Department.HOD
    -> FROM Student
    -> RIGHT JOIN Department
    -> ON Student.Dept_ID = Department.Dept_ID;
+------------+--------+-------+-----------------+------------+
| Student_ID | Name   | Marks | Department_Name | HOD        |
+------------+--------+-------+-----------------+------------+
|        201 | Arjun  |    81 | AIML            | Dr. Kumar  |
|        202 | Sneha  |    89 | ISE             | Dr. Reddy  |
|        203 | Kiran  |    74 | AIML            | Dr. Kumar  |
|        204 | Ananya |    91 | EEE             | Dr. Thomas |
|        205 | Vikram |    86 | ISE             | Dr. Reddy  |
|       NULL | NULL   |  NULL | MECH            | Dr. Patel  |
+------------+--------+-------+-----------------+------------+
6 rows in set (0.01 sec)
```

QUERY AND SUB QUERY:

```
mysql> USE CollegeStudentDB;
Database changed
mysql>
mysql> CREATE TABLE Student (
    ->     Student_ID INT PRIMARY KEY,
    ->     Name VARCHAR(50),
    ->     Department VARCHAR(20),
    ->     Marks INT,
    ->     City VARCHAR(30)
    -> );
Query OK, 0 rows affected (0.02 sec)

mysql>
mysql> INSERT INTO Student VALUES
    -> (301, 'Mohith KV', 'CSE', 88, 'Chennai'),
    -> (302, 'Ananya', 'IT', 91, 'Bangalore'),
    -> (303, 'Rahul', 'CSE', 75, 'Hyderabad'),
    -> (304, 'Divya', 'ECE', 86, 'Chennai'),
    -> (305, 'Karthik', 'IT', 94, 'Mumbai'),
    -> (306, 'Sneha', 'CSE', 71, 'Bangalore'),
    -> (307, 'Aditya', 'ECE', 82, 'Pune');
Query OK, 7 rows affected (0.01 sec)
Records: 7  Duplicates: 0  Warnings: 0

mysql>
mysql>
mysql> SELECT *
    -> FROM Student;
+------------+-----------+------------+-------+-----------+
| Student_ID | Name      | Department | Marks | City      |
+------------+-----------+------------+-------+-----------+
|        301 | Mohith KV | CSE        |    88 | Chennai   |
|        302 | Ananya    | IT         |    91 | Bangalore |
|        303 | Rahul     | CSE        |    75 | Hyderabad |
|        304 | Divya     | ECE        |    86 | Chennai   |
|        305 | Karthik   | IT         |    94 | Mumbai    |
|        306 | Sneha     | CSE        |    71 | Bangalore |
|        307 | Aditya    | ECE        |    82 | Pune      |
+------------+-----------+------------+-------+-----------+
7 rows in set (0.00 sec)
```

```
mysql>
mysql>
mysql> SELECT Name, Marks
    -> FROM Student
    -> WHERE Department = 'CSE';
+-----------+-------+
| Name      | Marks |
+-----------+-------+
| Mohith KV |    88 |
| Rahul     |    75 |
| Sneha     |    71 |
+-----------+-------+
3 rows in set (0.00 sec)

mysql>
mysql>
mysql> SELECT Name, Marks
    -> FROM Student
    -> WHERE City = 'Chennai'
    -> AND Marks > 80;
+-----------+-------+
| Name      | Marks |
+-----------+-------+
| Mohith KV |    88 |
| Divya     |    86 |
+-----------+-------+
2 rows in set (0.00 sec)

mysql>
mysql>
mysql> SELECT MAX(Marks) AS Highest_Marks
    -> FROM Student;
+---------------+
| Highest_Marks |
+---------------+
|            94 |
+---------------+
1 row in set (0.00 sec)
```

```
mysql>
mysql>
mysql> SELECT Name, Department, Marks
    -> FROM Student;
+-----------+------------+-------+
| Name      | Department | Marks |
+-----------+------------+-------+
| Mohith KV | CSE        |    88 |
| Ananya    | IT         |    91 |
| Rahul     | CSE        |    75 |
| Divya     | ECE        |    86 |
| Karthik   | IT         |    94 |
| Sneha     | CSE        |    71 |
| Aditya    | ECE        |    82 |
+-----------+------------+-------+
7 rows in set (0.00 sec)

mysql>
mysql>
mysql> SELECT Name, Marks
    -> FROM Student
    -> WHERE Marks > 80;
+-----------+-------+
| Name      | Marks |
+-----------+-------+
| Mohith KV |    88 |
| Ananya    |    91 |
| Divya     |    86 |
| Karthik   |    94 |
| Aditya    |    82 |
+-----------+-------+
5 rows in set (0.00 sec)
```

```
mysql>
mysql>
mysql> SELECT Name, Marks
    -> FROM Student
    -> WHERE Marks > (
    ->     SELECT Marks
    ->     FROM Student
    ->     WHERE Name = 'Mohith KV'
    -> );
+---------+-------+
| Name    | Marks |
+---------+-------+
| Ananya  |    91 |
| Karthik |    94 |
+---------+-------+
2 rows in set (0.00 sec)

mysql>
mysql>
mysql> SELECT Name, Department
    -> FROM Student
    -> WHERE Department = (
    ->     SELECT Department
    ->     FROM Student
    ->     WHERE Name = 'Mohith KV'
    -> );
+-----------+------------+
| Name      | Department |
+-----------+------------+
| Mohith KV | CSE        |
| Rahul     | CSE        |
| Sneha     | CSE        |
+-----------+------------+
3 rows in set (0.00 sec)
```

```
mysql> SELECT Name, Marks
    -> FROM Student
    -> WHERE Department = 'CSE'
    -> AND Marks > (
    ->     SELECT AVG(Marks)
    ->     FROM Student
    ->     WHERE Department = 'CSE'
    -> );
+-----------+-------+
| Name      | Marks |
+-----------+-------+
| Mohith KV |    88 |
+-----------+-------+
1 row in set (0.00 sec)

mysql>
mysql>
mysql> SELECT Name, Department
    -> FROM Student
    -> WHERE Department IN (
    ->     SELECT Department
    ->     FROM Student
    ->     WHERE City = 'Bangalore'
    -> );
+-----------+------------+
| Name      | Department |
+-----------+------------+
| Mohith KV | CSE        |
| Ananya    | IT         |
| Rahul     | CSE        |
| Karthik   | IT         |
| Sneha     | CSE        |
+-----------+------------+
5 rows in set (0.00 sec)
```

```
mysql>
mysql>
mysql> SELECT Name, Marks
    -> FROM Student
    -> WHERE Marks > ALL (
    ->     SELECT Marks
    ->     FROM Student
    ->     WHERE Department = 'ECE'
    -> );
+-----------+-------+
| Name      | Marks |
+-----------+-------+
| Mohith KV |    88 |
| Ananya    |    91 |
| Karthik   |    94 |
+-----------+-------+
3 rows in set (0.00 sec)

mysql>
```

```
mysql> SELECT AVG(Marks) AS Average_Marks
    -> FROM Student;
+---------------+
| Average_Marks |
+---------------+
|       83.8571 |
+---------------+
1 row in set (0.00 sec)

mysql>
mysql>
mysql> SELECT Name, Marks
    -> FROM Student
    -> WHERE Marks > (
    ->     SELECT AVG(Marks)
    ->     FROM Student
    -> );
+-----------+-------+
| Name      | Marks |
+-----------+-------+
| Mohith KV |    88 |
| Ananya    |    91 |
| Divya     |    86 |
| Karthik   |    94 |
+-----------+-------+
4 rows in set (0.00 sec)

mysql>
mysql>
mysql> SELECT Name, Marks
    -> FROM Student
    -> WHERE Marks = (
    ->     SELECT MAX(Marks)
    ->     FROM Student
    -> );
+---------+-------+
| Name    | Marks |
+---------+-------+
| Karthik |    94 |
+---------+-------+
1 row in set (0.00 sec)
```